AI Weather Wise

AI-Powered Weather Insights & Forecasting Automation

PROBLEM IDENTIFICATION

Weather tracking and decision-making can be fragmented and complex, forcing users to manually analyze multi-day forecasts, severe weather alerts, and climate statistics. Traditional applications present raw weather numbers without actionable insights, personalized recommendations, or plain-language summaries. The documented root causes include inefficient manual analysis of multi-variable forecasts, limited context for outdoor planning, lack of automated safety and travel advice during severe weather, and disjointed alert notification systems across multiple geographical locations.

PROPOSED IDEA

Build a secure RESTful backend service that aggregates real-time weather metrics, multi-day forecasts, and severe weather alerts while integrating Google Gemini AI to generate natural-language daily weather overviews, dynamic clothing/activity recommendations, and safety advisories. The selected concept is **AI Weather Wise**, using Node.js, Express.js, MongoDB with Mongoose ODM, JWT authentication, bcrypt password encryption, standard weather API integrations, and Google Gemini AI.

KEY IDEAS

- **Secure Authentication & User Preferences:** Registration and login using bcrypt password hashing and JWT authentication, supporting saved default locations and metric/imperial preference settings.
- **Location & Favorite Management:** Complete CRUD operations for managing saved cities, custom coordinates, and regional favorites with ownership and authorization controls.
- **Real-Time & Forecast Integration:** Aggregation of current weather metrics (temperature, humidity, wind speed, UV index, cloud cover) alongside multi-day forecast data.
- **AI-Powered Weather Summaries & Advisories:** Automated generation of human-readable weather overviews, outdoor activity ratings, clothing recommendations, and climate safety advice via Google Gemini.
- **Severe Weather & Alert Processing:** Ingestion and parsing of official meteorological warnings with automated AI summary creation for dangerous conditions.
- **Centralized Error Handling & Schema Validation:** Robust input validation using Mongoose schemas and uniform REST API error responses.
- **Modular MVC Architecture:** Separating models, controllers, routing, services, and database operations.

TARGET USERS & VALUE

The primary users are daily commuters, outdoor enthusiasts, event planners, travel coordinators, authenticated users, and public clients. AI Weather Wise simplifies complex meteorological data into clear, human-understandable narratives, automates personalized daily advisories, enhances safety awareness during critical weather events, and exposes a versatile API for web and mobile frontends.

SELECTED SCOPE

Authentication and authorization, saved locations and category/favorite management, real-time and multi-day weather ingestion, AI-powered weather summary and recommendation generation, schema validation, centralized error handling, MongoDB persistence, and JSON REST API responses. The detailed project document defines roles for platform administrators, enterprise operational viewers, authenticated users, and public consumers.

CORE TECHNOLOGY STACK

Layer	Technology
Runtime / Backend	Node.js, Express.js